

Requirement Analysis Document

Project Name:

Airport Runway Management Simulator

Course:

Software Development 1

Programming Language:

Java

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1. Introduction

1.1 Project Overview

Airport Runway Management Simulator is a Java-based software application designed to simulate airport runway operations. The system manages aircraft takeoff and landing requests, assigns runways, tracks runway status, and helps ensure safe and efficient airport traffic management.

1.2 Purpose

The purpose of this project is to develop a simulation system that demonstrates how airport runways are managed efficiently while applying software development and object-oriented programming concepts.

1.3 Objectives

- Simulate airport runway operations.

- Manage aircraft landing requests.
- Manage aircraft takeoff requests.
- Assign available runways.
- Monitor runway status.
- Prevent runway conflicts.
- Apply Object-Oriented Programming concepts.
- Provide an interactive user interface.

2. Functional Requirements

FR-1: Aircraft Registration

The system shall allow users to register aircraft information.

FR-2: Landing Request

The system shall allow aircraft to request landing permission.

FR-3: Takeoff Request

The system shall allow aircraft to request takeoff permission.

FR-4: Runway Assignment

The system shall assign available runways automatically.

FR-5: Runway Status Monitoring

The system shall display runway availability status.

FR-6: Queue Management

The system shall maintain landing and takeoff queues.

FR-7: Emergency Handling

The system shall prioritize emergency aircraft.

FR-8: Flight Information Display

The system shall display aircraft details and runway assignments.

FR-9: Traffic Simulation

The system shall simulate airport traffic operations.

FR-10: Exit System

The system shall allow users to close the application safely.

3. Non-Functional Requirements

NFR-1: Usability

The system should provide a simple and user-friendly interface.

NFR-2: Performance

The application should process runway assignments quickly.

NFR-3: Reliability

The system should operate without crashes during simulation.

NFR-4: Maintainability

The source code should follow object-oriented programming principles.

NFR-5: Portability

The application should run on any system supporting Java Runtime Environment (JRE).

4. System Requirements

Hardware Requirements

- Processor: Dual-Core Processor or Higher
- RAM: 4 GB or Higher
- Storage: 200 MB Free Space

Software Requirements

- Java Development Kit (JDK 17 or later)
 - Java Swing
 - NetBeans / IntelliJ IDEA
 - Git
 - GitHub
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5. Assumptions and Constraints

Assumptions

- Airport data is entered correctly.
- Users understand basic airport operations.

- Java Runtime Environment is installed.

Constraints

- The application will be developed using Java.
 - Project duration is limited according to course requirements.
 - Simulation will use predefined aircraft and runway data.
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6. Expected Outcome

The final software will simulate airport runway management operations by handling aircraft takeoff and landing requests, assigning runways, maintaining traffic queues, and displaying runway status. The application will provide an educational understanding of airport traffic management while demonstrating software development and object-oriented programming concepts.

Conclusion

Airport Runway Management Simulator is designed to demonstrate airport traffic control and runway management processes through a software simulation. The project will enhance understanding of software development, graphical user interface design, data management, and object-oriented programming concepts while providing an efficient airport operation simulation.